

Inference problems associated with simple linear regression

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LSM

Mean of Y for a given value of X

$$\mu_n = \beta_0 + \beta_1 X_n \quad X_n \text{ is specified}$$

$$\hat{\mu}_n = \hat{\beta}_0 + \hat{\beta}_1 X_n \quad \leftarrow \text{is unbiased for } \mu_n$$

$$\text{since } \hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} \text{ is unbiased for } \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

Question (1): Is $\hat{\mu}_n$ the "best" estimator for μ_n ?

Question (2): Is there any principle that leads to the following tests?

$$\text{Test of hypotheses } \begin{cases} H_0: \beta_0 = \beta_0^0 \text{ vs } H_A: \beta_0 \neq \beta_0^0 \\ H_0: \beta_1 = \beta_1^0 \text{ vs } H_A: \beta_1 \neq \beta_1^0 \\ H_0: \mu_n = \beta_0^0 + \beta_1^0 X_n \text{ vs } H_A: \mu_n \neq \beta_0^0 + \beta_1^0 X_n \end{cases}$$

Form of the test statistic in each case
(assuming $\varepsilon_i \sim N(0, 1)$)

$$\frac{\hat{\theta} - \theta^0}{s(\hat{\theta})} \underbrace{\text{under } H_0}_{t_{(n-2)}}$$

Q1) Is $\hat{\mu}_n = \hat{\beta}_0 + \hat{\beta}_1 X_n$ the best estimator of μ_n ?

$$MSE(\hat{\theta}) = E_{\theta}(\hat{\theta} - \theta)^2 = (\text{Bias}_{\theta} \hat{\theta})^2 + \text{Var}_{\theta} \hat{\theta}$$

$$MSE(\hat{\mu}_n) = \text{Var}_{(\beta, \sigma^2)}(\hat{\mu}_n) = \sigma^2 \begin{pmatrix} 1 \\ X_n \end{pmatrix}^T (X^T X)^{-1} \begin{pmatrix} 1 \\ X_n \end{pmatrix}$$

"Best" among all unbiased estimator ??

Such an estimator will be called UMVUE.

UMVUE = uniformly minimum variance unbiased estimator.

Ex 1: $\bar{X}$ UMVUE for μ if $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

Ex 2: $\hat{\mu}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h$ is the UMVUE for $\mu_h = \beta_0 + \beta_1 X_h$
if $Y_i \stackrel{ind}{\sim} N(\beta_0 + \beta_1 X_i, \sigma^2)$

If we simply assume that ϵ_i indep with mean 0 and variance σ^2 , then $\hat{\mu}_h$ is the best linear unbiased estimator (BLUE). i.e., if we restrict to those estimators of the form $\underline{l}^T \underline{Y}$ where $\underline{l} \in \mathbb{R}^n$, such that $E(\underline{l}^T \underline{Y}) = \mu_h = \beta_0 + \beta_1 X_h$ for all (β, σ^2)

$$\text{then } \text{Var}(\hat{\mu}_h) \leq \text{Var}(\underline{l}^T \underline{Y})$$

Notice that $\hat{\mu}_h = \begin{pmatrix} 1 \\ X_h \end{pmatrix}^T \hat{\beta}$ where $\hat{\beta}$ is the least squares estimator of β

$$\mu_h = \begin{pmatrix} 1 \\ X_h \end{pmatrix}^T \beta = \underline{a}^T \beta$$

What about the form of the BLUE?

Note: We estimate a linear function of β , viz.

$\underline{a}^T \beta$ by a linear function of $\underline{Y}$, i.e. $\underline{l}^T \underline{Y}$

$$\text{Unbiasedness} \Leftrightarrow \underline{a}^T \underline{\beta} = E(\underline{l}^T \underline{Y}) = \underline{l}^T E \underline{Y} = \underline{l}^T X \underline{\beta} \text{ for all } \underline{\beta} .$$

$$\Leftrightarrow \underline{a} = X^T \underline{l}$$

$$\hat{\mu}_h = \underline{a}^T \hat{\underline{\beta}} = \underbrace{\underline{a}^T (X^T X)^{-1} X^T}_{\underline{l}^T} \underline{Y}$$

$$\text{Problem } \underline{l}: \min_{X^T \underline{l} = \underline{a}} \text{Var}(\underline{l}^T \underline{Y}) \quad \{\text{exc.}\}$$

$$\underline{\sigma}^2 \underline{l}^T \underline{l}$$

Test for $H_0: \beta_1 = \beta_1^0$ vs. $H_A: \beta_1 \neq \beta_1^0$ — α

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

Claim 1: The t-test for α is equivalent to the likelihood ratio test

Claim 2: Suppose further that $\beta_1^0 = 0$ ^(WLOG). In that case, we can perform an alternative test that is based upon the notion of goodness of fit.

LRT $\Theta = \text{parameter space}$

$$\Theta = \Theta_0 \cup \Theta_A$$

$$LR = \frac{\sup_{\theta \in \Theta_0} f(Y|\theta)}{\sup_{\theta \in \Theta} f(Y|\theta)}$$

Goodness of Fit: Based on ANOVA

Compare $SSE_{full} \leq SSE_{reduced}$

↑ under H_0

If $\frac{SSE_{reduced} - SSE_{full}}{SSE_{full}}$ is large, then the parameter must have some impact

$$SSE_{full} = \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 = \sum e_i^2 = SSE$$

$$SSE_{reduced} = \sum (Y_i - \bar{Y})^2 = SST_0$$

ANOVA $SST_0 = SSR + SSE$

$$\frac{SSE_{reduced} - SSE_{full}}{SSE_{full}} = \frac{SST_0 - SSE}{SSE} = \frac{SST_0}{SSE} - 1$$

$$\frac{SSR}{SSE} = \frac{\hat{\beta}_1^2 \sum (X_i - \bar{X})^2}{(n-2) MSE}$$

one to one R^2

$$= \frac{1}{n-2} \left(\frac{\hat{\beta}_1 - 0}{\sqrt{\frac{MSE}{\sum (X_i - \bar{X})^2}}} \right)^2 = \frac{1}{n-2} t^2$$

$B(\hat{\beta}_1)$

Summary: large value of $|t| = \left| \frac{\hat{\beta}_1 - \beta_1^0}{s(\hat{\beta}_1)} \right|$

$\Leftrightarrow$ large value of R^2

$\Leftrightarrow$ large value of $\frac{SSE_{reduced} - SSE_{full}}{SSE_{full}}$

LR: $H_1 = \{(\beta_0, \beta_1, \sigma^2) : \beta_0, \beta_1 \in \mathbb{R}, \sigma^2 > 0\}$
 under $H_0: \beta_1 = 0$ $H_0 = \{(\beta_0, 0, \sigma^2) : \beta_0 \in \mathbb{R}, \sigma^2 > 0\}$

$$\sup_{\theta \in H_1} f(\underline{y}|\theta) = \sup_{(\beta_0, \beta_1, \sigma^2)} \frac{1}{(\sigma^2)^{n/2}} \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum (Y_i - \beta_0 - \beta_1 X_i)^2}$$

$\rightarrow$ maximizer is $(\underbrace{\hat{\beta}_0, \hat{\beta}_1}_{LS}, \underbrace{\tilde{\sigma}^2}_{\sum e_i^2/n})$

$$= \frac{1}{(2\pi)^{n/2} (\tilde{\sigma}^2)^{n/2}} e^{-n/2}$$

$$\sup_{\theta \in H_0} f(\underline{y}|\theta) = \sup_{(\beta_0, \sigma^2)} \frac{1}{(\sigma^2)^{n/2}} \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum (Y_i - \beta_1 X_i)^2}$$

$\rightarrow$ maximizer is $(\bar{Y}, \underbrace{\frac{1}{n} \sum (Y_i - \bar{Y})^2}_{\tilde{\sigma}_Y^2})$

$$= \frac{1}{(2\pi)^{n/2}} \frac{1}{(\tilde{\sigma}_Y^2)^{n/2}} e^{-n/2}$$

$$\Rightarrow LR = \left(\frac{\tilde{\sigma}^2}{\tilde{\sigma}_Y^2} \right)^{n/2} = \left(\frac{SSE}{SST_0} \right)^{n/2}$$

Rejecting for small values of LR

$\Leftrightarrow$ Rejecting for large values of $\left(\frac{SST_0}{SSE} - 1\right)$